

worked_example

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$y \sim X + 1 + (x|subject)$
 $y = X\beta + Z\mu + \epsilon$
 $X = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \\ 1 & 1 \\ 1 & 2 \\ 1 & 3 \\ 1 & 3 \end{bmatrix}$
 $Z = \begin{bmatrix} 1 & 1 & 0 & 0 & 0 & 0 \\ 1 & 2 & 0 & 0 & 0 & 0 \\ 1 & 3 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 2 & 0 & 0 \\ 0 & 0 & 1 & 3 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 & 1 & 3 \end{bmatrix}$
 $\beta = \begin{bmatrix} \beta_1 \\ \beta_2 \end{bmatrix}$
 $\mu = \begin{bmatrix} \mu_1 \\ \mu_2 \\ \mu_1 \\ \mu_2 \\ \mu_1 \\ \mu_2 \end{bmatrix}$
 $\epsilon \sim N(0, \sigma^2)$
 $n_{subject} = 3$
 $q = p \times n_{subject}$
 $u \sim N(0, G)$
 $G_{q \times q}$

$x_{n \times 1}$ is the independent variable from three different subjects, colour coded.

$y_{n \times 1}$ is the dependent variable, measured from the same subjects, colour coded.

$X_{n \times p}$ is the design matrix for the general effects across participants (first level effects) (intercept and slope)

$Z_{q \times p}$ is the design matrix for the effects for each participant (second level effects) (intercept and slope for each subject)

$\beta_{p \times 1}$ are the coefficients for the two parameters of X

$u_{q \times 1}$ are the coefficients for the six parameters of Z ($q = n_{subjects} \times p$)

$\epsilon_{n \times 1}$ are the residuals, assumed to normally distributed with a mean of 0 and a variance of σ^2

u is normally distributed with a mean of 0 and a variance-covariance structure $G_{q \times q} = uu^T$